

begin to paint a picture of how important native audio support is becoming.

Canvas and Video

“Your powers combine... I am Captain Planet!”

Together? Yes! This wonderful combination opens up an entirely new dimension for web. On a canvas, you have control over every pixel of the canvas. So when a video runs on a canvas, you have entire control over the video! Real time modifications can be made to the video. Various new effects can be added to the video like changing backgrounds, manipulating colours, breaking it up, putting together and anything at all.

In the you can see clicking the video causes it to fragment into pieces, each fragment still playing a part of the video!

Geolocation

The Geolocation API lets you find out where the user is and keep tabs on them as they move around, always with the user’s consent. This functionality could be used as part of user queries, e.g. to guide someone to a destination point. It could also be used for “geo-tagging” some content the user has created, e.g. to mark where a photo was taken.

The API is device-agnostic; it doesn’t care how the browser determines location, so long as clients can request and receive location data in a standard way. The underlying mechanism might be via GPS, wifi, or simply asking the user to enter their location manually. Since any of these lookups is going to take some time, the API is asynchronous; you pass it a callback method whenever you request a location.

At it’s most simple use, the Geolocation API looks like this:

```
• function get_location() {
•   navigator.geolocation.getCurrentPosition(show_map);
• }
```

HTML5Rocks.com has a great sample application that shows how to calculate distance travelled between two points. Check out http://www.html5rocks.com/en/tutorials/geolocation/trip_meter/ for more details.

Web storage and offline web apps

“Web” and “online” are two closely associated terms, downright synonymous to many people. So why on earth would we talk about “offline” web technologies, and what does the term even mean? HTML5 and related